🚕 Stakeholder Analysis NYC Taxi Trip Duration & Congestion Pricing Prediction

Primary Stakeholder: NYC Taxi Drivers and Small Fleet Operators

📋 Specific User Persona

Maria Santos is an independent NYC yellow cab driver with 8 years of experience. Since the implementation of CBD congestion pricing on **January 5, 2025**, her operational decisions have become more complex. Each ride request now requires her to quickly assess trip profitability by considering trip duration, fare amount, and the new congestion fee structure.

Maria's Operational Challenge

When Maria receives a ride request through her dispatch system or from a street hail, she must make a split-second decision: **accept or decline**. This decision directly impacts her hourly earnings. With congestion pricing, she needs to estimate:

1. **How long will this trip take?** (to maximize trips per hour and hourly earnings)
2. **What will the total fare be?** (including the congestion fee impact on passenger willingness)
3. **Is this trip profitable?** (balancing time investment against net earnings)

Pre-congestion pricing, Maria relied on experience and intuition. The new fee structure has added complexity: while the **$0.75 driver fee** is modest, congestion pricing affects passenger costs and demand patterns in CBD zones, influencing trip acceptance dynamics.

Why Existing Solutions Don't Address Maria's Needs

1. Taxi Fare Calculators

Examples: taxi-calculator.com, ride.guru

Limitation: Estimate passenger costs using distance-based formulas but don't predict trip duration for driver operational planning or provide ML-based predictions using actual trip patterns.

2. Historical Machine Learning Models

🔗 Kaggle NYC Taxi Trip Duration Competition (2016 data)

Link: https://www.kaggle.com/c/nyc-taxi-trip-duration

What it does: Predicts trip duration using 2016 trip data

Limitation: Uses data from 9 years before congestion pricing implementation. Cannot analyze fee patterns (the cbd_congestion_fee variable didn't exist) or address post-2025 operational realities.

🔗 Top-Performing GitHub Solution (Top 4% ranking, 2016 data)

Link: https://github.com/pklauke/Kaggle-NYCTaxi

What it does: Sophisticated feature engineering for duration prediction using 2016 data

Limitation: No congestion pricing context. Duration prediction only - no fee exposure analysis. Cannot address the new operational questions drivers face in 2025.

3. Navigation Apps

Examples: Google Maps, Waze

Limitation: Provide real-time traffic estimates but don't incorporate taxi-specific patterns, don't predict congestion fee exposure, and don't integrate duration and fee

for profitability analysis.

How Our Tool Serves Maria's Needs

🤖 Model 1: Trip Duration Prediction (Regression)

Input: Pickup zone, dropoff zone, hour, day of week, distance

Output: Predicted duration in minutes

Driver value: Estimate trip time before accepting to maximize hourly earnings

🏷️ Model 2: Congestion Fee Exposure Prediction (Classification)

Input: Pickup zone, time of day, day of week

Output: Binary prediction (Yes/No) with confidence score

Driver value: Understand total passenger cost, CBD zone demand patterns, and trip acceptance optimization

Maria's Decision-Making Workflow

📱 How Maria Uses Our Tool

❌ Scenario 1: Unprofitable Trip

📍 Ride Request (8:15 AM Tuesday)

Pickup: Times Square (CBD)

Dropoff: Upper East Side

Distance: 3.2 miles

🤖 Our Model Predictions:

Duration: 18 minutes

Fee Status: YES (95% confident)

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Maria's Quick Math:

Estimated fare: $16.00
Minus driver fee: -$0.75
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Net earnings: $15.25
Time: 18 minutes
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Hourly rate: ~$51/hour
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❌ DECLINE

Below $60/hour threshold

✅ Scenario 2: Profitable Trip

📍 Ride Request (8:45 AM Tuesday)

Pickup: Times Square (CBD)

Dropoff: JFK Airport

Distance: 16.5 miles

🤖 Our Model Predictions:

Duration: 52 minutes

Fee Status: YES (fee applies)

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Maria's Quick Math:

Estimated fare: $65.00
Minus driver fee: -$0.75
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Net earnings: $64.25
Time: 52 minutes
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Hourly rate: ~$74/hour
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✅ **ACCEPT**

Above threshold - Good trip!

💡 **Key Insight:** Same pickup location, same time, different destinations → Different profitability decisions based on our dual predictions!

What Makes Our Approach Novel

Our project offers novelty in two key dimensions:

1. Post-Congestion Pricing Implementation Data

All existing ML work uses pre-2025 data without congestion pricing:

🔗 **Kaggle NYC Taxi Trip Duration (2016 data):**
https://www.kaggle.com/c/nyc-taxi-trip-duration

Uses data from 9 years before congestion pricing existed. Cannot predict fee exposure because the cbd_congestion_fee variable didn't exist in the dataset.

🔗 **Top GitHub Solution (Top 4%, 2016 data):**
https://github.com/pklauke/Kaggle-NYCTaxi

Sophisticated feature engineering but uses 2016 data with focus purely on duration prediction. No congestion fee analysis possible.

✨ **Our Contribution:** We're using `January 2025 data` with actual **cbd_congestion_fee** values — providing machine learning analysis based on real taxi trips after congestion pricing was implemented, capturing how the new fee structure changed travel and demand patterns.

2. Dual-Model Operational Support

Existing models predict duration OR cost in isolation. We uniquely combine:

- **Primary value:** Duration prediction (regression) helps drivers maximize earnings by optimizing trips/hour
- **Secondary value:** Fee exposure prediction (classification) provides context on:
  - Total passenger cost estimation
  - CBD zone demand patterns
  - Trip acceptance optimization

While the $0.75 per-trip driver fee is modest, understanding fee patterns helps drivers navigate the new operational landscape where passenger costs and demand

have shifted due to congestion pricing.

Success Metrics

📊 Technical Performance Goals:

- **Regression model:** RMSE < 10 minutes for duration prediction
- **Classification model:** Accuracy > 85% for fee exposure prediction

💼 Operational Value:

- Models provide actionable insights for driver trip acceptance decisions
- Interpretable feature importance identifies key factors affecting duration and fee exposure
- Dual-model integration enables informed profitability assessment

References

📚 Key Sources

1. **Kaggle NYC Taxi Trip Duration Competition (2016 data):** https://www.kaggle.com/c/nyc-taxi-trip-duration

2. **Top GitHub Solution - pklauke (Top 4%, 2016 data):**
   https://github.com/pklauke/Kaggle-NYCTaxi